

경북대학교 SRT 현황



강민찬, 강태호, 김재영 (경북대학교)



한국연구재단
National Research Foundation of Korea

2023년 8월 22일 전파사용자 회의
서울대학교 시흥캠퍼스



A small telescope for beginners of radio astronomy & university educations

- Increasing importance of radio astronomy in recent days, despite its difficulties for beginners and university students
- Small radio telescopes in individual universities and institutes can greatly help for education purpose
 - Several initiating projects such as the MIT Haystack's SRT (also several in Korea)
- A lot of small research projects can also be performed; from imaging the sun, flux monitoring, spectral line measurements (esp. HI), finding and tracking variable continuum radio sources

CONTACT NEWS EVENTS MEMOS INTERNAL SEARCH G

About ▼Our Researchers ▼Astronomy ▼Geodesy ▼Geospace ▼Space Technology ▼

Home · Education & Public Outreach · SRT: The Small Radio Telescope

SRT: The Small Radio Telescope

The Small Radio Telescope (SRT) was developed by Haystack Observatory to serve as an educational tool for universities and colleges for teaching astronomy and radio technology.

NOTE: The SRT software has been newly overhauled and updated—please [see below for the latest version](#).

The original SRT was popular, with several hundred being built and is still in use at many colleges. It is no longer available new, however, and advances in electronics have rendered the original obsolete. Development of a new SRT was undertaken by Haystack in 2012.

The primary philosophy of the latest design for the SRT is that the users should build the telescope themselves from commercially available equipment, based on plans, instructions, and software provided by Haystack. It is designed to be assembled easily with minimal need for special tools or skills. This approach provides an educational opportunity to the users who build the telescope. Also, users of the SRT will be more familiar with how it works if they assemble it themselves, allowing better in-house technical assistance.

The SRT is no longer available as a kit from Cassi Corp. If you decide to build an SRT, we recommend using the TV dongle version of the New SRT using parts list provided below.

[The SRT wiki site](#)

[SRT Memo Series](#)

SRT assembly information

- Hardware manual ([pdf](#))
- Parts list ([pdf](#))



Characteristic of our small radio telescope

- SPIDER 230C model from Radio2Space (a company based in Italy)
- Aluminum dish with $3 \times 4\text{mm}$ holes (The Parabolic Mesh Type Dish)
- Diameter : 2.3m
- 1420 MHz radio astronomy receiver, 50 MHz instantaneous bandwidth radiometer and 1024 channels spectrometer (61 KHz spectral resolution)



Our small radio telescope



Receiver and Feed Horn



The Feed Horn



Low Noise Amplifier



The Receiver

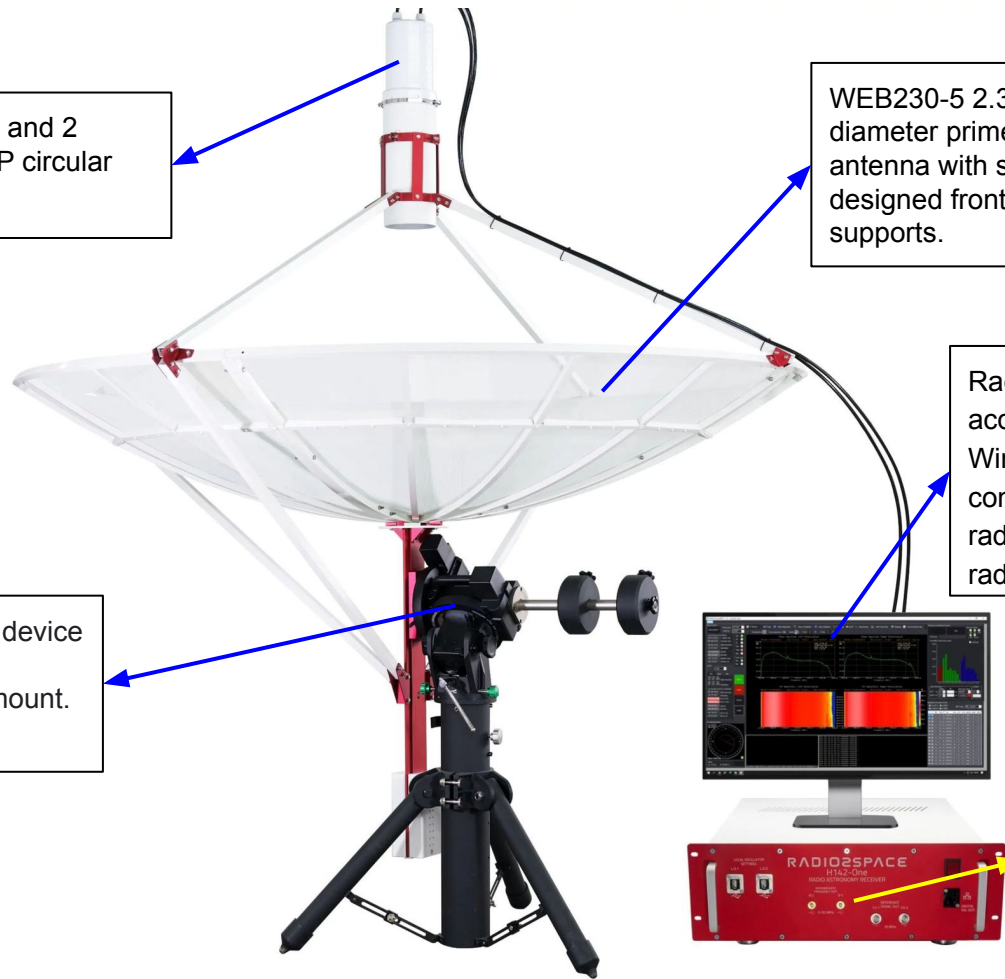
1420 MHz optimised feed and 2 LNAs for LHCP and RHCP circular polarisation

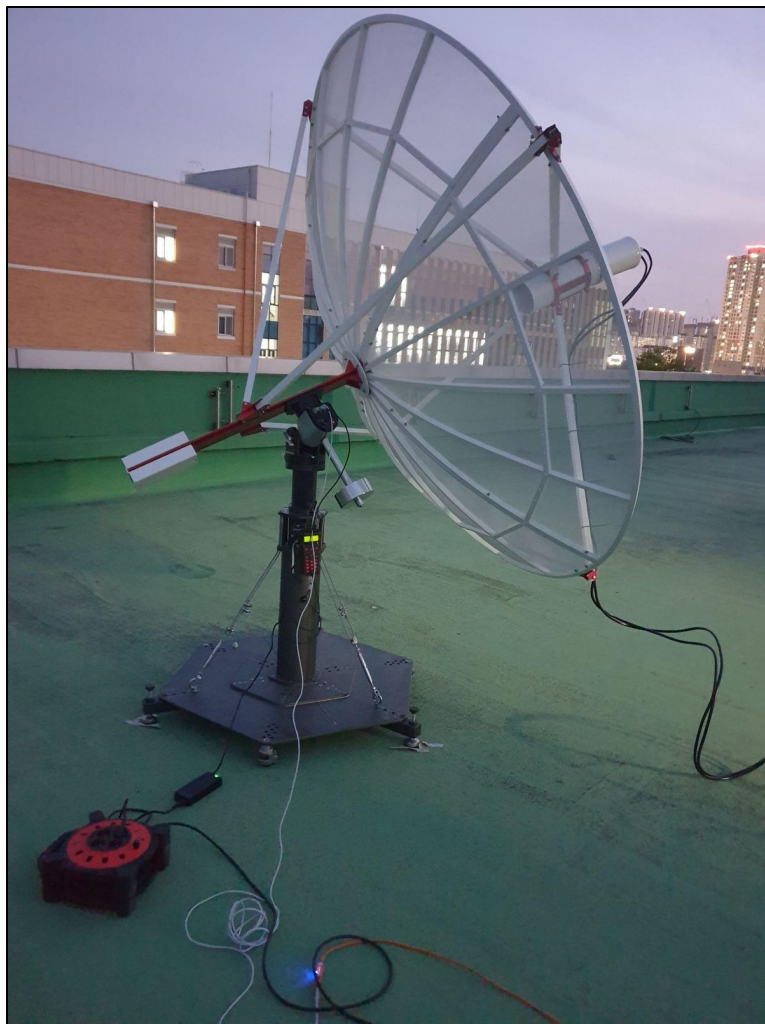
WEB230-5 2.3 meter diameter prime focus antenna with special designed front and rear supports.

RadioUniversePRO control and acquisition software for Windows 10/11, designed to control all devices of SPIDER radio telescope and to collect radio astronomy data.

Equatorial mounts support device allows to use the parabolic antenna on an equatorial mount.

H142-One 1420 MHz radio astronomy receiver, 50 MHz instantaneous bandwidth radiometer and 1024 channels spectrometer to be installed in the control room on a standard 19" rack or on a table.





The Mount

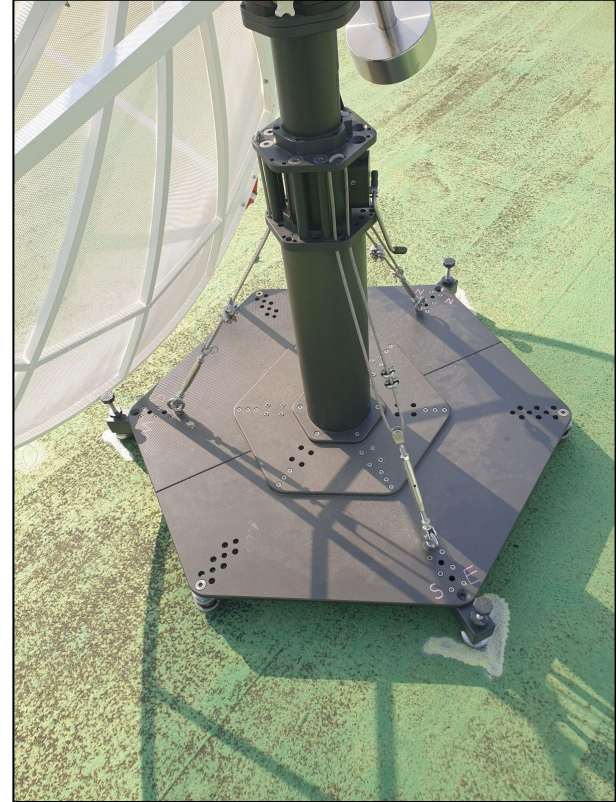


- **RST-300** Weightless Mount of RainbowAstro
- Payload : 30kg (Maximum 50kg)
- Harmonic Drive : No Backlash
- Made in Korea (i.e. prompt and easy A/S)

Tent



Observing Procedure : Move the Telescope



Observing Procedure : Connect the Cables

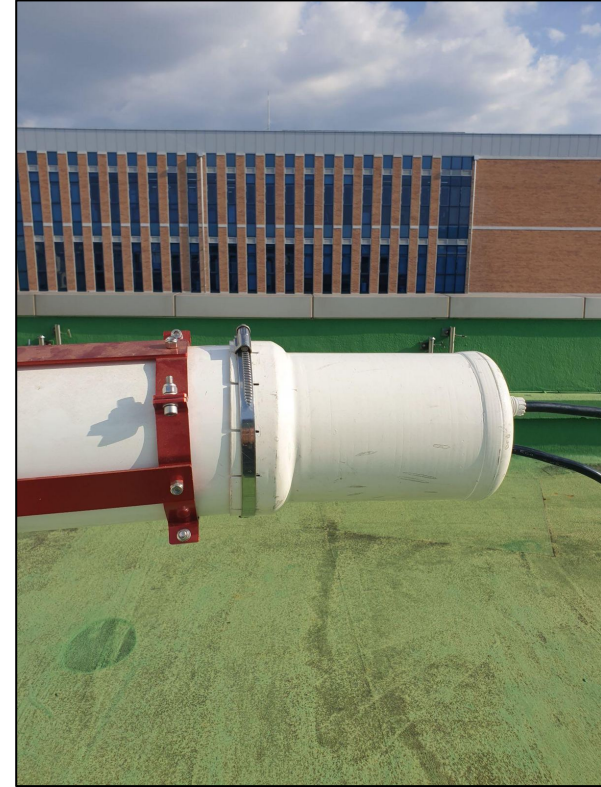


Connect

- the Power
- Hand Controller
- Computer

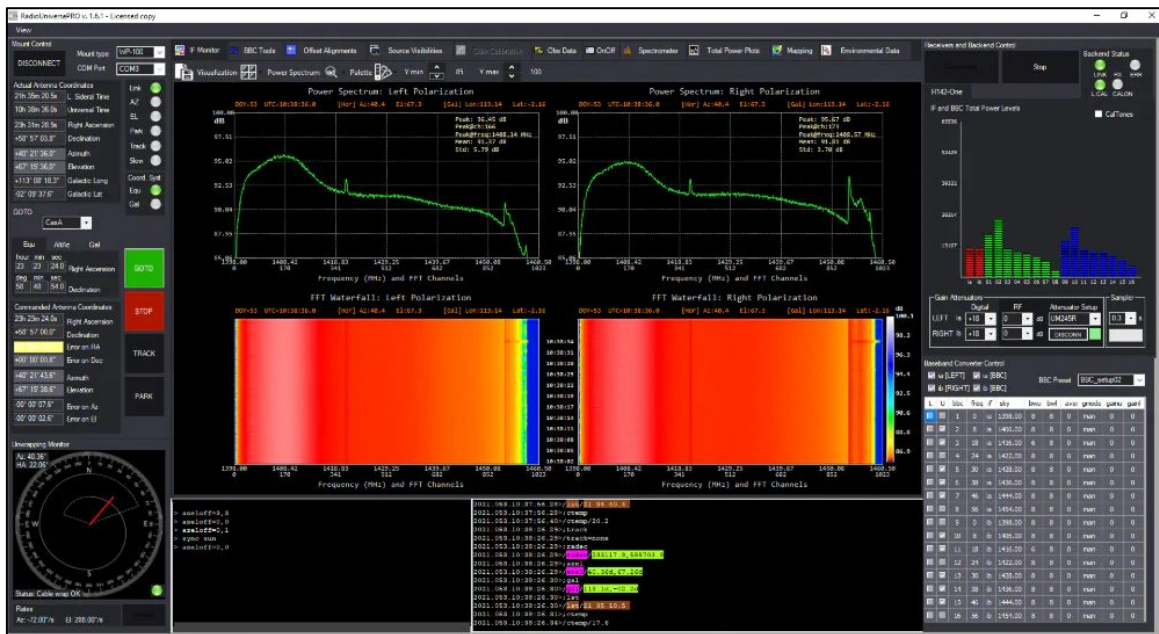
to the Mount (RST-300)

Observing Procedure : Connect the LNA



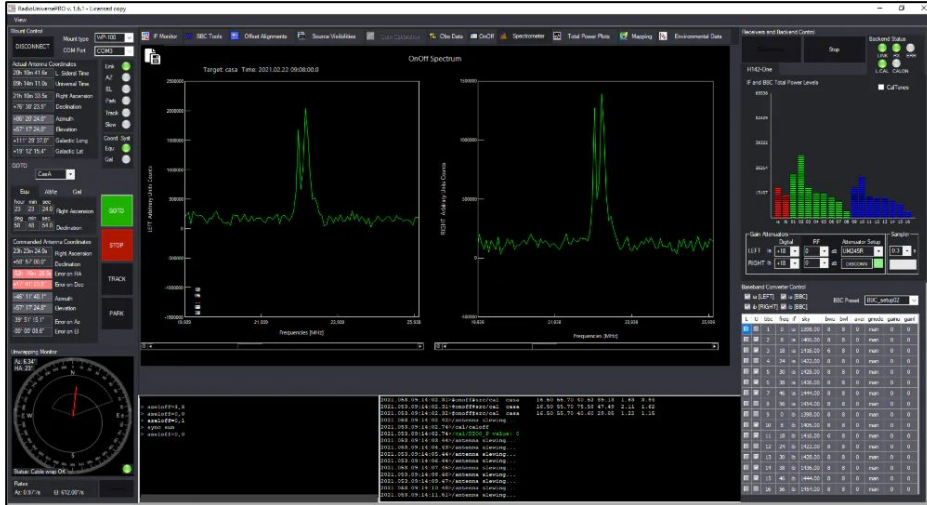


The Software : RadioUniversePro

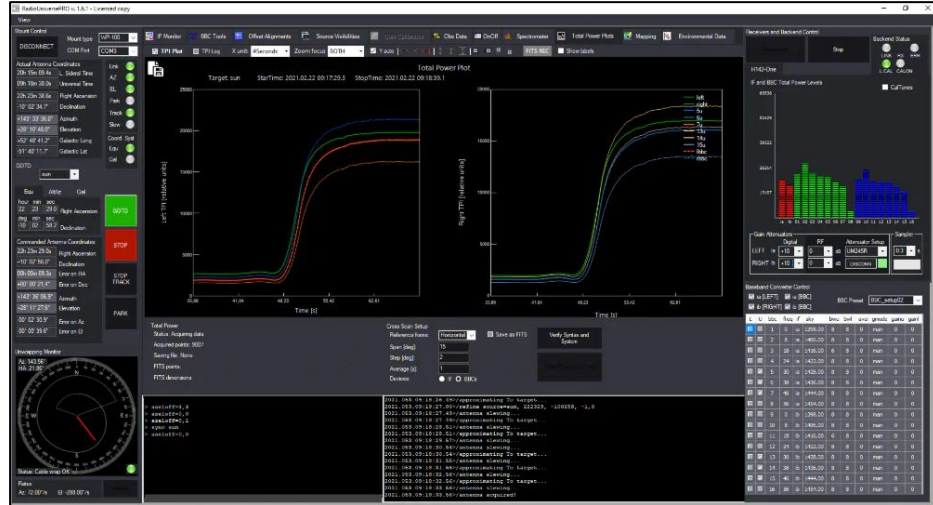


- Control Software for SPIDER Radio Telescope
- IF Monitor, Cross Scan, BBC Tool, OnOff, Spectrometer, Total Power Plots, Mapping, etc..

Example Screen of RadioUniversePro

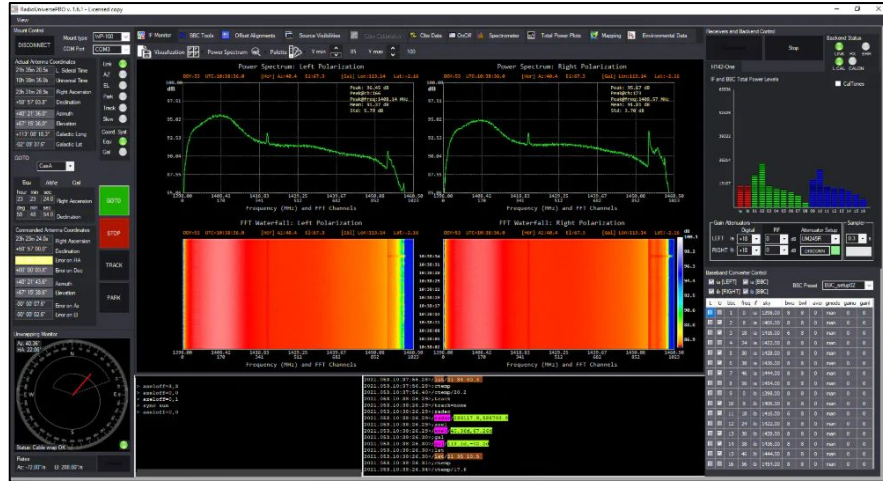


Spectrometer Mode

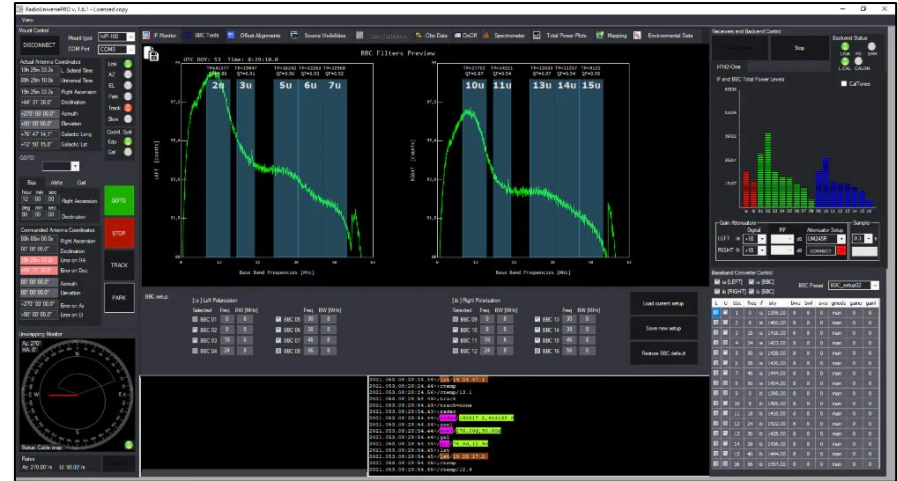


Total Power Plots Mode

Example Screen of RadioUniversePro

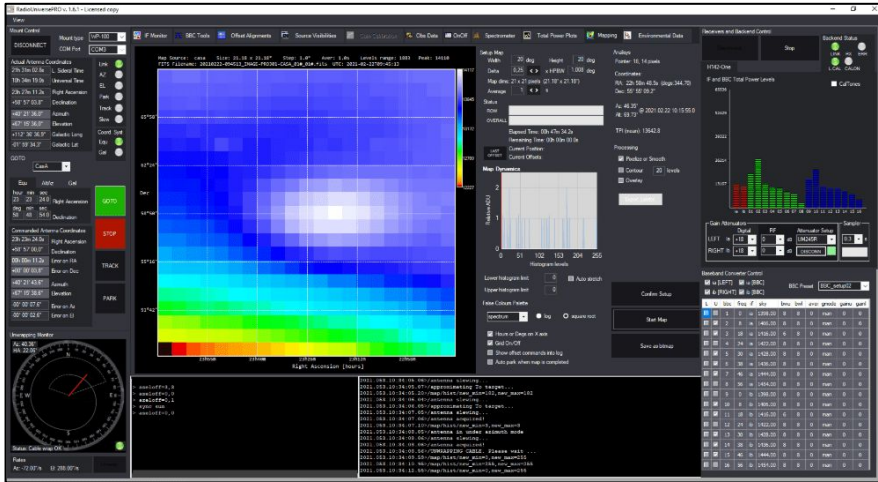


IF Monitor

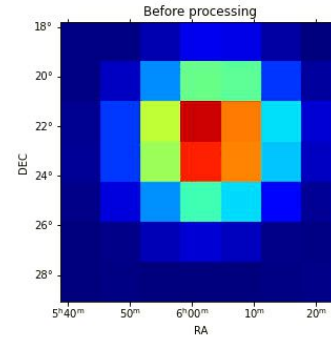


BBC Tool

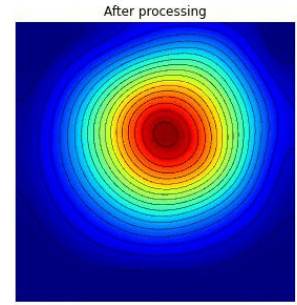
Example Screen of RadioUniversePro



Mapping



OBJECT: sun
TELESCOP: Spider 300A
BEAM: 4.03 deg
OBSFREQ: 1429.25 MHz
SPECSYS: LSRK
POLARIZ: L/R



BANDWIDTH: 62.50 MHz x 2
DATE-OBS: 2020-06-21T05:02:40
DATE-END: 2020-06-21T05:16:44
SITE: TAI TAM, HONG KONG

by courtesy of the Stanley Ho Astronomical Observatory and Hong Kong Astronomical Society

Calibration scheme / System temperature measurement

- On-Off Observation

On : A commercial black sponge (assuming as a room temperature black body)

Off : Empty Sky

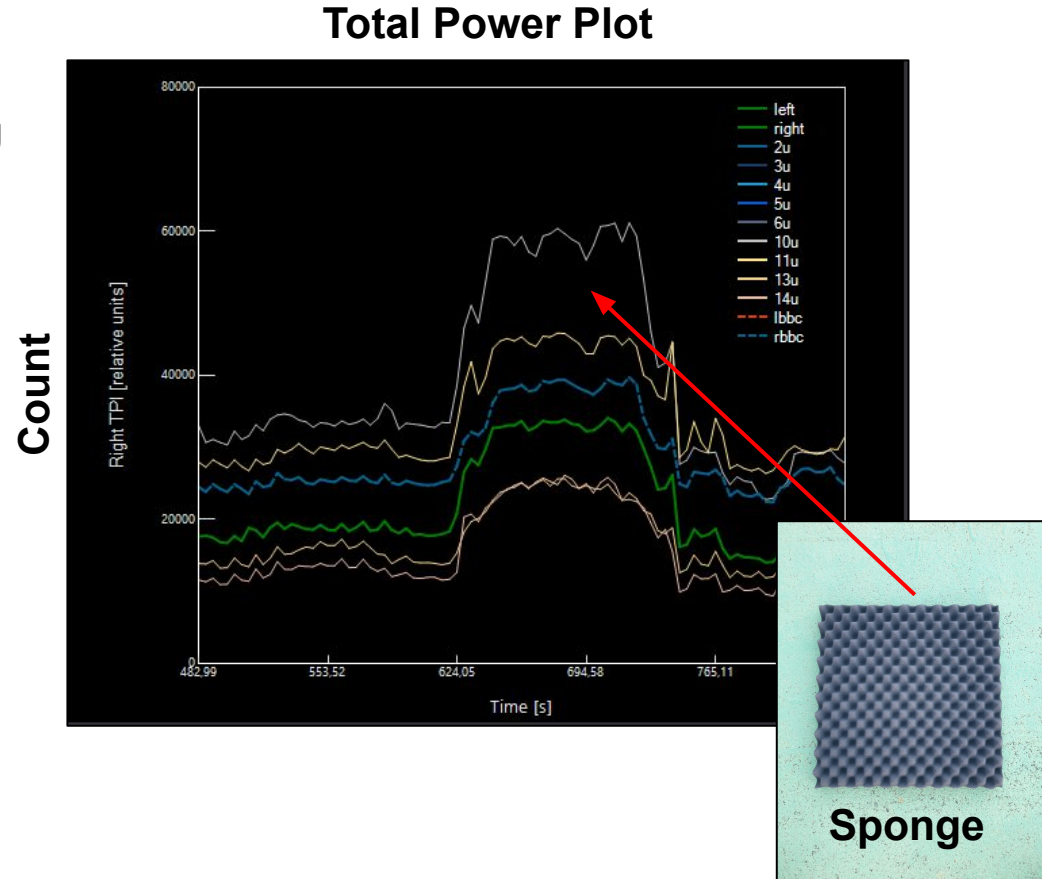
- Y-factor Measurement

Hot Count ~59000 (Sponge)

Cold Count ~32000

$Y = \text{Hot Count} / \text{Cold Count}$

$= 59000 / 32000 = 1.84$



Tsys Calculation

- System Temperature

$$Y = 1.84$$

$$T_{\text{cold}} = T_{\text{MW}} + T_{\text{Atm}} + T_{\text{CMB}} + \dots \sim 10 \text{ K}$$

$$T_{\text{hot}} = 273.15 + 35 \text{ K} = 308.15 \text{ K}$$

An Introduction To Radio Astronomy (Burke) **Eq 6.10**

$$\mathbf{T_{sys} \sim 346K}$$

$$T_{sys} = \frac{T_{\text{hot}} - YT_{\text{cold}}}{Y - 1}$$

cf. MIT SRT (Located in Haystack)

SRT $T_{\text{sys}} = 150\text{K}$

GNU $T_{\text{sys}} = 346\text{K}$

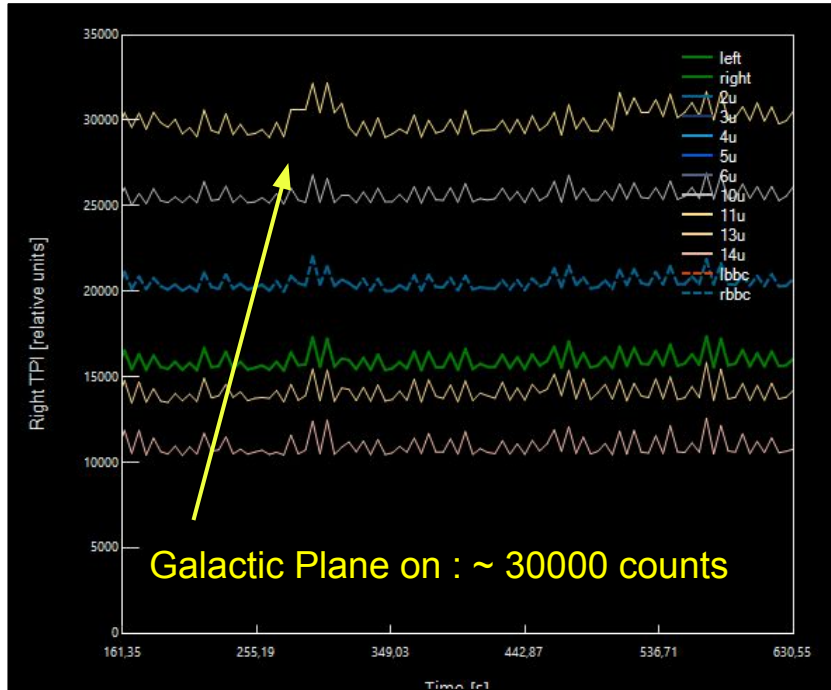
1] Characteristics

L.O. Frequency range	1370-1800 MHz
L.O. Tuning steps	40 kHz
L.O. settle time	<5 ms
Rejection of LSB image	>20 dB
3 dB bandwidth	56 kHz
I.F. center	40 kHz
6 dB I.F. range	10-70 kHz
Preamplifier frequency range	1400-1440 MHz
Typical system temperature	150 K
Typical L.O. leakage out of preamp input	-105 dBm
Preamplifier input for dB compression from out of band signals	-24 dBm
Preamplifier input for intermodulation interference	-30 dBm
Square law detector approx. max.	2000 K at 0 dB attenuation 20,000 K at 10 dB attenuation
Control	RS-232 2400 baud

Considering the newer and lower-noise radio receiving systems employed for the 230C, our higher T_{sys} likely indicates the presence of higher radio interference noise in the city.

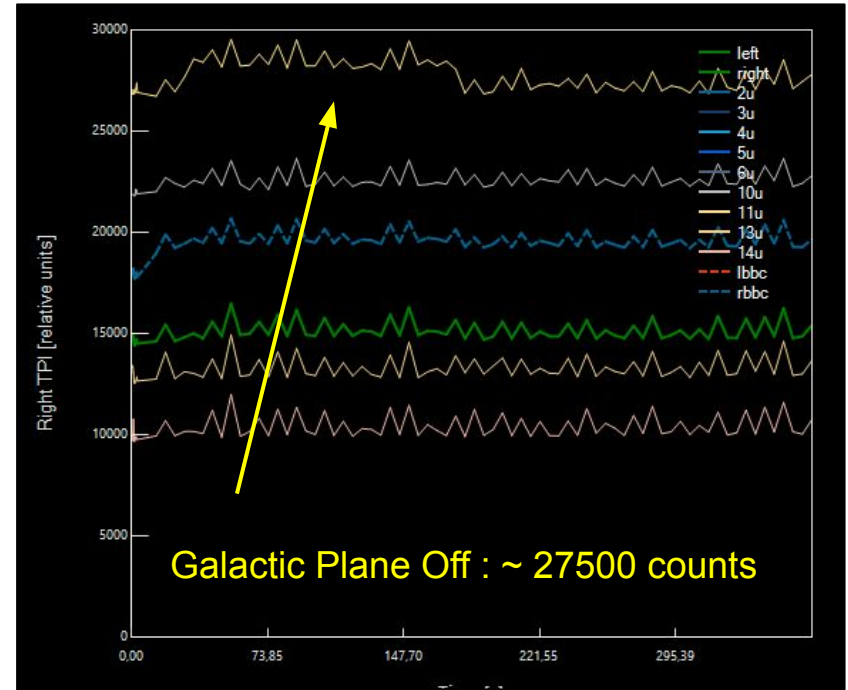
ref. SRT Memo

Example : Galactic Plane



$l = \sim 4^\circ$, $b = \sim 3^\circ$

X-axis : Time
Y-axis : Counts



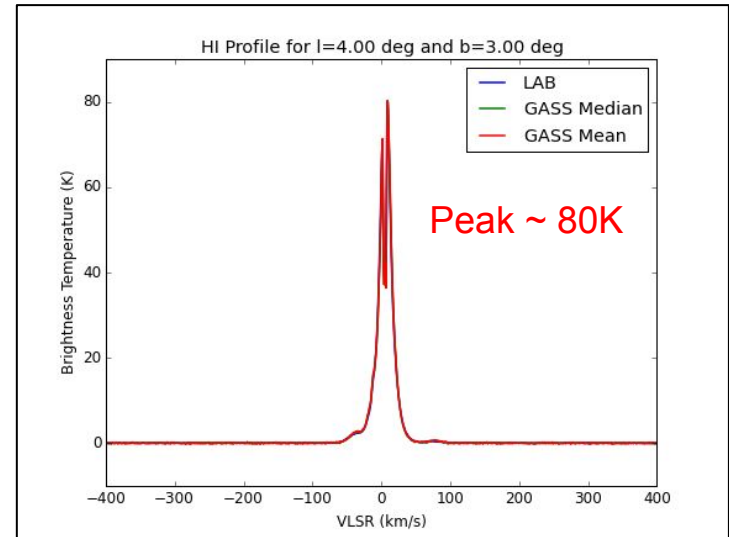
$\sim$ Zenith

Example : Galactic Plane

- Single Dish Calibration Techniques at Radio Wavelengths (K. O'Neil) **Eq. 20**

$$T_{\text{Source}} = \left[\frac{T_{\text{ON}} - T_{\text{OFF}}}{T_{\text{OFF}}} \right] T_{\text{off}}$$

- Ratio Term = $(30000 - 27500) / 27500 = \sim 0.091$
- $T_{\text{off}} \sim T_{\text{sys}} = 346\text{K}$
- $T_{\text{source}} = 31.5\text{K}$ (Average)
- We can also use the known line temperatures to determine the aperture efficiency of the telescope.



Leiden/Argentine/Bonn (LAB) Galactic HI survey

Future Plan

- Short-term:
 - Improving on the maintenance and operation
 - Training observers (typically students so far)
 - Continued system verification
 - Quantifying pointing errors, aperture efficiency determination, absolute flux calibration, ...
- Long-term:
 - Pipelining the telescope operation and data processing
 - Mapping the Galactic plane HI
 - Fainter continuum sources (by position switching mode)
 - Finding and monitoring variables/transients
 - Open telescope to users from local and outside communities

Conclusion

- We have successfully introduced the new radio telescope
- Early tests show promising results, including the line and continuum sources (so far Galactic HI and the Sun)
- We have faced several technical issues, from which students learned a lot about the radio astronomy and practical aspects of operating an observatory
- **A great facility to motivate students and to gain experience of radio astronomy!**